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Thermodynamic modelling of Y–H and Y–Zr–H system aided by first-principles and its application in bulk hydride moderator fabrication

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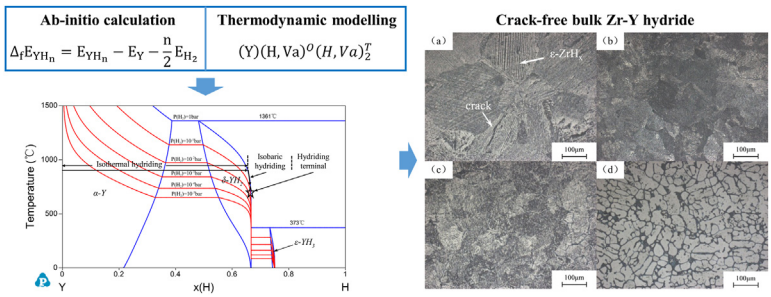
HIGHLIGHTS

- A self-consistent thermodynamic description of Y–H and Y–Zr–H system was derived.
- The ab-initio enthalpies of formation were adopted for thermodynamic assessment.
- A three-stage hydriding process was established based on thermodynamic calculation.
- The martensitic transformation from δ to ε zirconium hydride is inhibited by yttrium.
- Zirconium hydride without micro-cracks can be fabricated with addition of yttrium.

GRAPHICAL ABSTRACT



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ABSTRACT

A self-consistent thermodynamic description of Y–H binary system has been derived using the CALPHAD method with the aid of first-principles calculations and applied in the fabrication of crack-free bulk Zr–Y hydride neutron moderator. The enthalpies of formation of the end-members calculated by first-principles were adopted for the thermodynamic assessment. Only four model parameters were needed and optimized with the available experimental data from literature. The calculated phase equilibria and thermodynamic data agree well with experimental data. Y–Zr–H ternary system was constructed by extrapolation of three binary sub-systems. According to the thermodynamic calculation results, the hydrogenation of Zr–Y alloys was divided into three stages: isothermal hydriding, isobaric hydriding and terminal hydriding. By carefully controlling the temperature and hydrogen pressure, the transformation from metal to hydride of yttrium and zirconium can be separated and carried out in different stages, so that the cracks formation during hydrogenation was successfully inhibited. With addition of yttrium, the stress induced by the martensitic transformation from δ to ε zirconium hydride would be relaxed. As a result, hydride embrittlement can be mitigated, and the bulk Zr–Y hydrides without micro-cracks can be fabricated.

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1. Introduction

There has been an increasing interest in the rare earth-hydrogen systems because of their energy-related applications such as hydrogen storage [1–4], hydrogen sensing [5–7] and nuclear industry [8]. Recently, much attention has been attracted by yttrium-hydrogen (Y–H) system. Yttrium hydride can be served as catalyst of Mg-based hydrogen storage material to improve its sluggish hydrogen absorption/desorption kinetics by forming an 18R-type long-period stacking ordered (LPSO) phase nanocomposite [9–11]. Reversible change from yellowish transparent YH_3 to shiny reflected YH_2 upon exposure to hydrogen gas allows application in eye-visible hydrogen sensor [12,13]. The oxygen-containing yttrium hydride films have emerged as promising candidates for the next generation of energy-efficient smart windows owing to its reversible photochromic properties at ambient conditions [14,15]. Furthermore, because yttrium hydride exhibits high hydrogen density and low hydrogen desorption pressure, it is used as neutron moderator in compact high-temperature nuclear reactors [16,17]. For these applications, it is essential to gain a fundamental knowledge on the thermodynamics of the Y–H system.

In our recent work, it was demonstrated that yttrium is able to resist hydrogen loss from the zirconium hydride at the reactor working temperature. Hence, Zr–Y hydride is superior to conventional zirconium hydride and proposed as a potential candidate for neutron moderator in compact nuclear reactor [18,19]. Cracks would be easily formed during hydrogenation of metal and dramatically affect the moderating performance and integrity of the hydride. However, it is a formidable challenge to fabricate the crack-free bulk Zr–Y hydride because of the stress induced by the phase transformation from metal to hydride and the brittleness of the formed hydride. Futhermore, the significantly large gap of equilibrium hydrogen pressures between the zirconium and yttrium makes the task almost unconquerable. It is desirable to probe the hydrogen absorption process from the thermodynamic perspective.

Recently, Y–H system has been experimentally studied and assessed by Fu et al. [20]. However, this thermodynamic description can be further improved to give less and more reliable model parameters. In this study, a self-consistent thermodynamic description of Y–H binary system is derived using the CALPHAD method with the aid of first-principles calculations. To investigate the hydrogenation process of Zr–Y alloys, Y–H, Zr–H, and Y–Zr binary systems were combined and extrapolated to construct the Y–Zr–H ternary system. Based on the thermodynamic calculation, the crack-free bulk Zr–Y hydrides have been successfully fabricated and characterized. The mechanism of zirconium hydride embrittlement mitigation by yttrium has been established. This provides a typical approach to utilize the thermodynamics of metal-hydrogen system to technological applications.

2. Literature survey

The phase diagram of yttrium-hydrogen system was reported by Khatamian and Manchester in 1988 [21] and assessed recently by Fu et al. in 2018 [20]. This system is composed of a gas phase, liquid phase and four solid phases. The solid phases are a metallic phase α-Y and two nonstoichiometric hydride phases: δ-YH_2 and ε-YH_3. The other allotropic form of metallic yttrium, β-Y, and the liquid phase are not considered in our work, because their temperature ranges are much higher than the hydrogenation temperature [21]. The α-Y phase is a solid solution with hcp_A3 structure. The hydrogen atoms mainly occupy the tetrahedral sites [22] and can dissolve into the α-Y lattice up to 21 at%, which is the solubility limit of hydrogen [23]. The dihydride phase δ-YH_2 has a CaF_2 type fcc_C1 crystal structure. For this hydride phase, the hydrogen atoms first locate in the tetrahedral sites, and then occupy the octahedral sites when the theoretical stoichiometric ratio is exceeded [24]. The trihydride phase ε-YH_3 has a HoD_3 type hexagonal-close-packed (hcp) crystal structure. The neutron diffraction technique shows that the 18 interstitial hydrogen atoms occupy the tetrahedral sites (12g), octahedral sites (4d) and triangular sites (2a) of the ε-YH_3 hydride unit cells [25]. The homogeneity ranges are determined to be $YH_{1.9}$-$YH_{2.2}$ and $YH_{2.8}$-YH_3 at room temperature for δ-YH_2 and ε-YH_3, respectively [26,27]. The structural information of all phases in the Y–H binary system is summarized in Table 1.

Pressure composition isotherms (PCI) measurement is the most informative and accurate experimental method to acquire phase equilibria information of rare earth-hydrogen system. Phase equilibria relations and thermodynamic data such as invariant reaction equilibria, phase boundaries, solubility limits and plateau pressures can be obtained from PCI measurements. The PCI curves of Y–H system at various temperatures were determined by Lundin and Blackledge [28], Yannopoulos et al. [29], Begun et al. [30], Tanase et al. [31], Chernikov et al. [32] and Fu et al. [20]. However, only Lundin's group reported the PCI data above 1000 °C. It should be noted that the equilibrium temperature (449 °C) of the invariant reaction between YH_2-YH_3 phases determined by Fu et al. [20] using PCI measurement was not consistent with that extrapolated from the DSC measurements under different hydrogen pressures, which was 369 °C. Therefore, the PCI measurement results of Fu et al. [20] at low temperatures are not included in the thermodynamic modelling. The PCI results, especially phase boundaries, differ from each other a little because of the different gaseous impurities levels of the raw materials and precision of the Sieverts apparatus.

Other experimental techniques can also be used to study the phase equilibria and thermodynamic properties of the Y–H system. Beaudry et al. [23] and Bonnet et al. [33] determined the solubility limits of hydrogen in α-Y at temperatures ranging from 250 °C to 850 °C and from 130 °C to 400 °C, respectively, using X-ray diffraction technique. The YH_2-YH_3 phase boundary at room temperature was measured by Nuclear Magnetic Resonance technique [34]. Moreover, Dantzer and Kleppa [35] and Akhachinskii [36] determined the standard enthalpy of formation of yttrium hydride using calorimetric method. The testing results were consistent with the data calculated from the PCI measurements. The detailed experimental data such as invariant reactions, PCI, solubility branches and plateau pressures from the literatures are listed in the Table 2.

3. Thermodynamic modelling

The CALPHAD (CALculation of PHase Diagrams) method is an efficient semi-empirical technique to give a self-consistent thermodynamic description of the rare earth-hydrogen system. In the thermodynamic database, the Gibbs energies of all the phases involved in the given system are described by the so-called compound-energy formalism (CEF) as a function of temperature and composition with adjustable parameters using suitable thermodynamic models [37].

3.1. Pure elements

The Gibbs energies of the pure elements such as Y in hcp and fcc structure and hydrogen in the gas phase with respect to the

Table 1
The structural information of the phases in the Y–H binary system.

Phase	Space group	Prototype	Homogeneity Range (at%)	Lattice parameters		Reference
α-Y	P63/mmc	Mg	0–21	a = 3.6636	c = 5.7900	[23]
δ-YH_2	Fm3$\overline{m}$	CaF_2	65.5–68.8	a = 5.2070		[24]
ε-YH_3	P3$\overline{c}$1	HoD_3	73.7–75	a = 6.360	c = 6.6590	[25]

Table 2
List of the experimental data of the literature.

Type of data	Composition range	Temperature range (°C)	Reference	Data used for modelling
Invariant reaction	GAS(H_2)+HCP(Y)→δ-YH_2	1366	[28]	Yes
	GAS(H_2)+δ-YH2→ε-YH_3	363[a]	[29]	Yes
PCI	α-Y–δ-YH_2	900<T < 1350	[28]	Yes
	α-Y–δ-YH_2	600<T < 950	[29]	No
	α-Y–δ-YH_2	700<T < 1000	[30]	No
	α-Y–δ-YH_2	500<T < 888	[31]	No
	α-Y–δ-YH_2	600<T < 1000	[32]	No
	α-Y–δ-YH_2	650<T < 900	[20]	Yes
	δ-YH_2–ε-YH_3	250<T < 350	[20]	No
	δ-YH_2–ε-YH_3	250<T < 350	[29]	Yes
Solubility branch	α-Y–δ-YH_2	250<T < 850	[23]	Yes
	α-Y–δ-YH_2	130<T < 400	[33]	No
Plateau pressure	α-Y–δ-YH_2	900<T < 1350	[28]	Yes
	α-Y–δ-YH_2	600<T < 950	[29]	No
	δ-YH_2–ε-YH_3	250<T < 350	[29]	Yes
	α-Y–δ-YH_2	650<T < 900	[20]	Yes
	δ-YH_2–ε-YH_3	250<T < 350	[20]	No
	α-Y–δ-YH_2	500<T < 1000	[30–32]	No

[a] This transformation temperature is extrapolated to 1 bar using the average value of the absorption and desorption plateau pressures.

reference state are taken from the Scientific Group Thermodata Europe (SGTE) database [38]. The molar Gibbs energy of the element i in the Φ phase can be expressed as a function of the temperature as below:

$$^{0}G_i^{\varphi} = G_i^{\varphi} - H_i^{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where H_i^{SER} is the molar enthalpy of the element in the stable state at 298.15 K and 100 kPa, and a to h are the coefficients from the database.

3.2. Gas phase

The gas phase is constituted by the species H and H_2. The molar Gibbs energy of the gas phase can be described by the ideal solution model as the following equation:

$$G_m^{gas} = \sum_i x_i[^{0}G_i^{gas} + RT\ln(P_i/P_0)] \quad (2)$$

where $^{0}G_i^{gas}$ is the molar Gibbs energy of the pure gaseous specie i in the standard state and x_i is the molar fraction of the specie i in the gas phase. P_i is the partial pressure of the specie i and P_0 is the standard reference pressure (100 kPa).

3.3. Solid phases

For phases α-Y and δ-YH_2, the neutron diffraction analysis shows that the H atoms are preferentially located in the two tetrahedral sites and one octahedral site. For the hcp HoD_3 structure ε-YH_3, two-thirds of the H atoms occupy the tetrahedral sites, and the remaining one-third of the H atoms are located for 2/3 in the octahedral sites and for 1/3 in the triangular sites, forming a trigonal structure [39]. Because this trigonal structure of ε-YH_3 phase is fairly similar to that of α-Y phase, we consider them as the same phase with different hydrogen concentration. Thus the ε-YH_3 phase can be modelled using the same sublattice as α-Y. All the solid phases in the Y–H binary system can be described using only one same three-sublattice model $(Y)(H,\ Va)^{O}(H,\ Va)_2^{T}$ with O referring to the octahedral sites and T to the tetrahedral sites. When the H atoms mainly occupy the tetrahedral interstitial sites and the octahedral interstitial sites remain vacant, the three-sublattice model represents the α-Y phase. On the contrary, when the H atoms occupy almost all the octahedral and tetrahedral interstitial sites, the model represents the ε-YH_3 phase. As a result, the molar Gibbs energies of the solid phases FCC_C1 (δ-YH_2) and HCP_A3 (α-Y and ε-YH_3) can be calculated using the following equation:

$$G_m^{\varphi} = {}^{ref}G_m^{\varphi} + {}^{id}G_m^{\varphi} + {}^{ex}G_m^{\varphi} \quad (3)$$

where $^{ref}G_m^{\varphi}$, $^{id}G_m^{\varphi}$ and $^{ex}G_m^{\varphi}$ represent the reference, ideal and excess Gibbs energy of the phase Φ, respectively.

The reference molar Gibbs energy $^{ref}G_m^{\varphi}$ can be expressed as below:

$$^{ref}G_m^{\phi} = y'_H y''_H G^{\phi}_{Y:H:H} + y'_H y''_{Va} G^{\phi}_{Y:H:Va} + y'_{Va} y''_H G^{\phi}_{Y:Va:H} + y'_{Va} y''_{Va} G^{\phi}_{Y:Va:Va} \quad (4)$$

where y'_i and y''_i refer to the site fraction of the element i in the second and third sublattice, respectively. $G^{\phi}_{Y:H,Va:H,Va}$ represents the Gibbs free energy of various end members in the Φ phase with different site occupation of H atoms.

The ideal molar Gibbs energy $^{id}G_m^{\varphi}$ can be expressed as below:

$$^{id}G_m^{\phi} = RT(y'_H\ln y'_H + y'_{Va}\ln y'_{Va}) + 2RT(y''_H\ln y''_H + y''_{Va}\ln y''_{Va}) \quad (5)$$

The excess molar Gibbs energy $^{ex}G_m^{\varphi}$ can be expressed as below:

$$^{ex}G_m^{\phi} = y'_H y'_{Va}(y''_H\,{}^0L^{\phi}_{Y:H,Va:H} + y''_{Va}\,{}^0L^{\phi}_{Y:H,Va:Va}) + y''_H y''_{Va}(y'_H\,{}^0L^{\phi}_{Y:H,Va:H} + y'_{Va}\,{}^0L^{\phi}_{Y:H,Va:Va}) \quad (6)$$

where ${}^0L^{\phi}_{Y:H,Va:*}$ and ${}^0L^{\phi}_{Y:*:H,Va}$ refer to interaction parameters between H and vacancy in the second and third sublattice, respectively. Such parameters can be described using a Redlich–Kister polynomial, and only the zero order interaction parameter was employed in this work. The interaction parameter is usually linearly dependent with temperature:

$$L^{\phi} = a + bT \quad (7)$$

where a and b are the adjustable parameters.

4. First-principles calculations

For the end-members of the Y–H system, except for the stable phases α-Y, δ-YH_2 and ε-YH_3, the other end-members are metastable. Hence it is difficult to measure the thermodynamic properties of the metastable end-members by the experimental techniques, making the thermodynamic optimization difficult to converge. In this work, the enthalpy of formation of the metastable end-members in the Y–H system were calculated using first-principles calculations, providing reliable data for the thermodynamic database construction.

The Vienna Ab-initio Simulation Package (VASP) was used to calculate the total energy at 0 K [40]. The interaction between electrons and ions was described by the Projector-Augmented Wave (PAW) method, and the exchange and correlation energy was estimated using the generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof functional (PBE) [41–43]. To ensure the precision of calculation, the energy cut-off was set to 400 eV, the Gamma-centered k-points in the Brillouin zone were generated by the Monkhorst-Pack procedure, and the energy convergence criterion for electronic self-consistency was 1×10^{-6} eV per atom.

The supercell was constructed according to the crystal structure and site occupation of the end-members as described above. The supercell was then optimized to obtain the relaxed structure. The total energy vs. volume (E-V) of each end-member was calculated based on the relaxed supercell using VASP. The derived E-V curves were fitted by the 4-parameter Birch-Murnaghan (BM) equation of state (EOS) [44], thus obtaining the 0 K total energy E_0, equilibrium volume V_0, bulk modulus B_0 and its first derivate with respect to pressure B_0'. All the calculated equilibrium properties of the end-members in the Y–H system were listed in the Table 3. The calculated results for fcc-YH_2 are in good agreement with the literature [45–47].

The 0 K total energy of the H_2 molecule was estimated to be −6.677 eV by putting the dimer into a 10 × 11 × 12 box, considering the spin polarization of the electrons. The calculated result is consistent with that reported by Kresse [48].

The formation enthalpies of the end-members can be calculated using the total energies of the end-members, hcp-Y and H_2 dimer as the following equation:

$$\Delta_f E_{YH_n} = E_{YH_n} - E_Y - \frac{n}{2}E_{H_2} \quad (8)$$

where E_{YH_n}, E_Y and E_{H_2} represent the 0 K static total energy of YH_n, hcp-Y and H_2, respectively. The enthalpy of formation and relaxed structural parameters of each end-member in the Y–H system are presented in the Table 4. The calculated enthalpies of formation of fcc-YH_2 and trigonal-YH_3 were in line with those reported by Dantzer and Kleppa [35] and Akhachinskii [36]. The optimized lattice parameters of the stable phases hcp-Y, fcc-YH_2 and trigonal-YH_3 were in accordance with the experimental data as listed in Table 1. Table 4 also shows the atomic positions of Y and tetrahedral, octahedral and triangular H in the end-members.

Table 3
The calculated equilibrium properties of end-members in the Y–H system.

Structure	Compound	E_0 (eV)	V_0 (Å^3)	B_0 (GPa)	B'_0
fcc	YH	−42.19843	121.72715	64.57	3.59095
	YH_2	−61.33918	142.10121	81.44	3.8007
	YH_3	−74.93142	138.60494	95.60	3.75499
hcp	Y	−12.90751	65.89224	40.1	3.13519
	YH	−21.09835	63.63158	62.26	3.31518
	YH_2	−28.98948	72.80172	72.23	3.74757
Trigonal	YH_3	−114.08256	231.71059	74.76	3.2867

5. Results and discussion

5.1. Thermodynamic assessment of Y–H binary system

The CALPHAD method consists of modelling the Gibbs energies of all the phases with appropriate thermodynamic models, fitting the adjustable parameters in the models with a critical set of experimental information, yielding a self-consistent thermodynamic database of the given system. The experimental information used for optimization includes phase equilibria and thermodynamic data, together with the theoretical prediction, for instance, first-principles calculations. In this work, the optimization was carried out using the Parrot module of the Thermo-Calc software [49].

The thermodynamic quantities calculated from the static total energies at 0 K were adopted for the thermodynamic assessment. The enthalpies of formation of the end-members were taken as the initial guess from the first-principles calculations as listed in Table 4. The entropies of formation of the stable compounds (fcc-YH_2 and hcp-YH_3) were taken from the literature [26]. On the other hand, the entropies of formation of the metastable end-members can be estimated by the entropy of the hydrogen gas, because the entropies of the solid phases can be neglected compared with the much larger entropy of the gas phase. Thus the entropy of formation per hydrogen atom was considered to be half of the entropy of formation of hydrogen molecule dimer (−65 J mol^{-1} K^{-1}) [50].

Considering the validity and consistency of the experiments and the temperature ranges, the PCI data measured by Lundin and Blackledge (1000 °C–1350 °C) [28], Yannopoulos et al. (251 °C–326 °C) [29] and Fu et al. (700 °C–900 °C) [20] as well as the solubility limits measured by Beaudry and Spedding [23] were included in the optimization. The phase boundaries of the δ-YH_2 - ε-YH_3 two-phase region were assessed using the PCI data of Fu et al. [20] rather than those of Yannopoulos et al. [29]. There exist contradictory reports in the literature about the value that should be taken as the thermodynamic equilibrium value (absorption, desorption or something in between). For RE-H system, the thermodynamic equilibrium hydrogen pressures should be an intermediate value between absorption and desorption [50]. Thus the average value of the absorption and desorption equilibrium pressures measured by Yannopoulos et al. [29] were used for thermodynamic optimization for δ-YH_2 - ε-YH_3 equilibria. For the other temperatures, only the absorption isotherms were adopted because the desorption isotherms are very difficult to measure due to long equilibrium time for RE-H systems. Whether the experimental data were used for modelling or not was presented in the last column of

Table 4
Enthalpies of formation and structural parameters of end-members in the Y–H system.

Structure	Compound	ΔH (kJ/mol)	Lattice parameters (Å)		Atomic positions			
			a	c	Element	x	y	z
fcc	YH	−73.0734	4.95037	–	H1(Octahedral)	1/2	1/2	1/2
	YH_2	−212.6575	5.20697 (5.20701 [24])	–	H1(Tetrahedral)	1/4	1/4	1/4
	YH_3	−218.4041	5.16199	–	H1(Octahedral)	1/2	1/2	1/2
					H2(Tetrahedral)	1/4	1/4	1/4
hcp	Y	0	3.6631 (3.6636 [23])	5.8147 (5.7900 [23])	Y	1/3	2/3	1/4
	YH	−73.0316	3.64125	5.50043	H1(Octahedral)	0	0	0
	YH_2	−131.6045	3.59237	6.39469	H1(Tetrahedral)	1/3	2/3	5/8
Trigonal	YH_3	−245.5073	6.33391 (6.360 [25])	6.59997 (6.6590 [25])	Y	0.66299	0	1/4
					H1(Tetrahedral)	0.35052	0.02883	0.09218
					H2(Octahedral)	1/3	2/3	0.18537
					H3(Triangular)	0	0	1/4

Table 2.

The invariant reaction, solubility limits as well as phase boundaries were first optimized. Then, the plateau pressures of various temperatures were taken into account to build the initial shape of the phase diagram. The PCI data were subsequently optimized, and eventually the plateau pressures were fitted for the second time to obtain the model parameters. In the metal-hydrogen system, the phase transition points between two phases (i.e. phase boundaries) are determined by the initial and terminal points of the hydrogen pressure plateaus. However, the PCI curve is composed of many interrupted measurement points, thus the measured phase boundaries are not that accurate. Therefore, our optimization relied more on the PCI data rather than the phase boundaries, and a lower weight was assigned to the phase boundary data in this work. For simplicity, the interaction parameters of atoms in the same lattice were assumed identical. The optimized Gibbs energies of all end-members and model parameters are presented and compared with Fu et al. [20] in Table 5. Compared with the Y–H system assessed by Fu et al. [20], this work is based on the thermodynamic data calculated by first-principles calculations, which is closer to physical reality, thus only four model parameters were required for optimization.

The phase diagram and property diagrams were calculated using the Pandat software, since it is powerful in calculating property contour diagrams in multi-component systems [51]. The calculated phase diagram of Y–H system under 1 bar hydrogen pressure is shown in Fig. 1 and compared with experimental data from the literature. It can be seen from the figure that the calculated phase diagram agrees well with the experimental phase boundaries determined by PCI measurements.

Nevertheless, as we interpreted above, the determination of the phase transition points was relatively subjective, leading to the error of the experimental phase boundaries. In this sense, the consistency of the PCI data between the calculation and the experiments is highly critical for the assessment. Fig. 2 shows the calculated PCI curves of various temperatures. The experimental PCI data, especially the desorption isotherms at 326 °C, 301 °C, 251 °C, are shown with the literature. The open symbols represent the PCI data upon absorption, while the solid ones represent the desorption data. We can see that the calculated PCI curves exhibit fairly good agreement with the experimental data used for modelling. The calculated PCI curves of δ-YH_2 - ε-YH_3 are between the experimental absorption and desorption isotherms. The PCI curves in the temperatures ranging from 251 to 326 °C are related to the phase transition from δ-YH_2 to ε-YH_3, while those in the temperatures ranging from 600 to 1300 °C are related to the phase transition from α-Y to δ-YH_2.

The equilibrium hydrogen content variation with the temperature at the constant pressure, i.e. isobars, under various hydrogen pressures were calculated based on our Y–H thermodynamic database aided by the Pandat software, as shown in Fig. 3. The isobars together with the phase diagram are very useful in predicting the hydrogen content at any given temperatures under a certain pressure, which is of great importance in the process of metal hydrogenation.

The gas phase related invariant reactions and phase boundaries can be seen from the calculated Y–H phase diagram. Table 6 lists the equilibrium temperature and hydrogen content of participant phases of the invariant reactions in the Y–H system under 1 bar hydrogen pressure. Compared with literature data, the calculated results are consistent with the PCI measurements of Lundin and Blackledge [28] and Yannopoulos et al. [29]. In the case of the equilibria between δ-YH_2 and ε-YH_3, the transformation temperature is calculated to be 373 °C. This value is between the transformation temperatures extrapolated from the absorption (349 °C) and desorption (390 °C) plateau pressures at various temperatures. It should be noted that the equilibrium temperature of the invariant reaction between YH_2-YH_3 phases is also close to the DSC

Table 5
The Gibbs energies and optimized parameters.

Phase	Parameters	
	Fu et al. [20]	Present work
fcc	$G^{FCC_C1}_{Y:H:Va} = G^{SER}_{Y} + G^{SER}_{H} - 47457.31 + 65 \times T$	$G^{FCC_C1}_{Y:H:Va} = G^{SER}_{Y} + G^{SER}_{H} - 73073.38 + 65 \times T$
	$G^{FCC_C1}_{Y:Va:H} = G^{SER}_{Y} + 2G^{SER}_{H} - 219675.52 + 136.5 \times T$	$G^{FCC_C1}_{Y:Va:H} = G^{SER}_{Y} + 2G^{SER}_{H} - 220832.88 + 141.93 \times T$
	$G^{FCC_C1}_{Y:H:H} = G^{SER}_{Y} + 3G^{SER}_{H} - 238524.26 + 195 \times T$	$G^{FCC_C1}_{Y:H:H} = G^{SER}_{Y} + 3G^{SER}_{H} - 218404.13 + 195 \times T$
	${}_0L_{Y:H,Va:*} = -10274.14\,{}_0L_{Y:*:H,Va} = 27422.43 - 25.16 \times T$	${}_0L_{Y:Va:H,Va} = 21209.85 - 21.57 \times T$
hcp	$G^{HCP_A3}_{Y:H:Va} = G^{SER}_{Y} + G^{SER}_{H} - 65831.08 + 65 \times T$	$G^{HCP_A3}_{Y:H:Va} = G^{SER}_{Y} + G^{SER}_{H} - 73031.65 + 65 \times T$
	$G^{HCP_A3}_{Y:Va:H} = G^{SER}_{Y} + 2G^{SER}_{H} - 169184 + 130 \times T$	$G^{HCP_A3}_{Y:Va:H} = G^{SER}_{Y} + 2G^{SER}_{H} - 131604.50 + 130 \times T$
	$G^{HCP_A3}_{Y:H:H} = G^{SER}_{Y} + 3G^{SER}_{H} - 252188.05 + 202.73 \times T$	$G^{HCP_A3}_{Y:H:H} = G^{SER}_{Y} + 3G^{SER}_{H} - 260881.58 + 208.71 \times T$
	${}_0L_{Y:H,Va:H} = -94404.77\,{}_0L_{Y:H,Va:Va} = -74238.15 + 5.1823 \times T$	${}_0L_{Y:*:H,Va} = -106287.38 + 9.75 \times T$

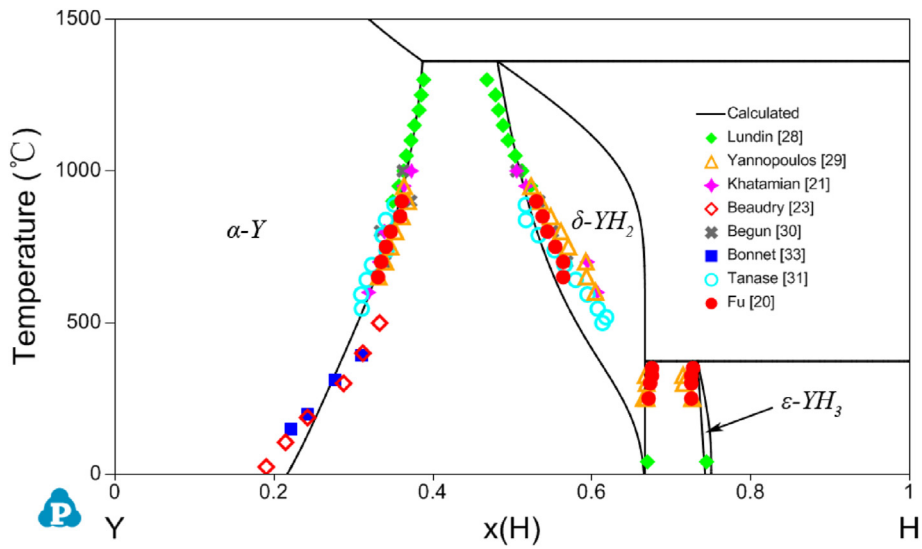

Fig. 1. The calculated Y–H phase diagram compared with experimental data.

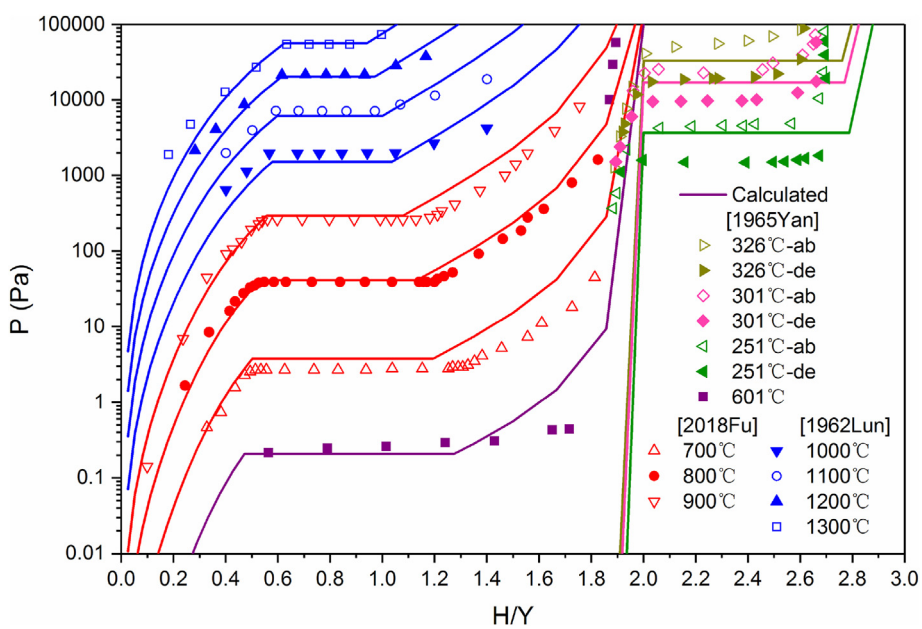

Fig. 2. The calculated PCI curves and experimental data from the literature.

measurements of Fu et al. [20]. It can be seen from the calculated phase diagram that the nonstoichiometric ε-YH_3 phase region exhibits a narrow homogeneity domain, which is in accordance with the experiments but not illustrated by the experimental phase diagram assessed by Khatamian and Manchester [21].

According to the phase law, the hydrogen pressure remains steady as the hydrogen content changes in the two-phase region during absorption or desorption of hydrogen gas. The unchanged equilibrium hydrogen pressure, namely pressure plateau, obeys van't Hoff equation. Thus thermodynamic data like the enthalpy and entropy of formation can be determined from the pressure plateau of different temperatures measured by PCI. The calculated equilibrium hydrogen pressures as a function of temperature in the two-phase regions of α-Y+δ-YH_2 and δ-YH_2+ε-YH_3 follow the linear van't Hoff equation, as illustrated in Fig. 4. The open squares represent the PCI data upon absorption, while the solid ones represent the desorption data. The calculated results are in good agreement with the experimental results. The calculated van't Hoff plot of δ-YH_2+ε-YH_3 is in the middle of the plateau pressure data of the absorption and desorption isotherms. Note that the equilibrium hydrogen pressures vs temperatures between δ-YH_2 and ε-YH_3 of Fu's calculation [20] and experimental data exhibit a large error between the Yannopoulos's experimental results [29] and our calculation. The phase regions where the stable phase exists are denoted in the figure. The points where the van't Hoff plots intersect with the 1 bar pressure line correspond to the phase transformation temperatures between the respective two phases, which are consistent with the equilibrium temperatures of the invariant reactions as listed in Table 6.

Fig. 5 presents the calculated enthalpies of formation at 25 °C in the Y–H binary system as a function of the hydrogen concentration. The calculated data exhibit good agreement with the calculated

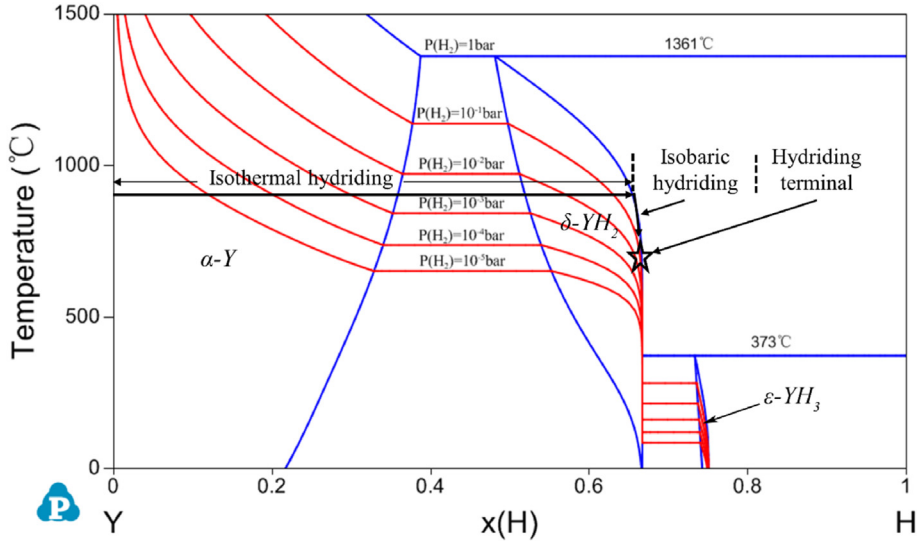

Fig. 3. The calculated isobars of Y–H system under various hydrogen pressures and the schematic diagram of hydriding stages.

Table 6
Invariant reactions in the Y–H system under 1 bar hydrogen pressure.

Reaction	Temperature (°C)	x(H)			Reference
GAS(H_2)+HCP(Y)→δ-YH_2	1361	0.387	0.481	1	This work
	1366	0.389	0.466	1	Lundin [28]
	1321	0.376	0.483	1	Fu [20]
GAS(H_2)+δ-YH2→ε-YH_3	373	0.667	0.733	1	This work
	349	0.669	0.714	1	Yannopoulos [29]-ab
	390	0.674	0.706	1	Yannopoulos [29]-de
	449	0.676	0.727	1	Fu [20]-PCI
	369	–	–	–	Fu [20]-DSC

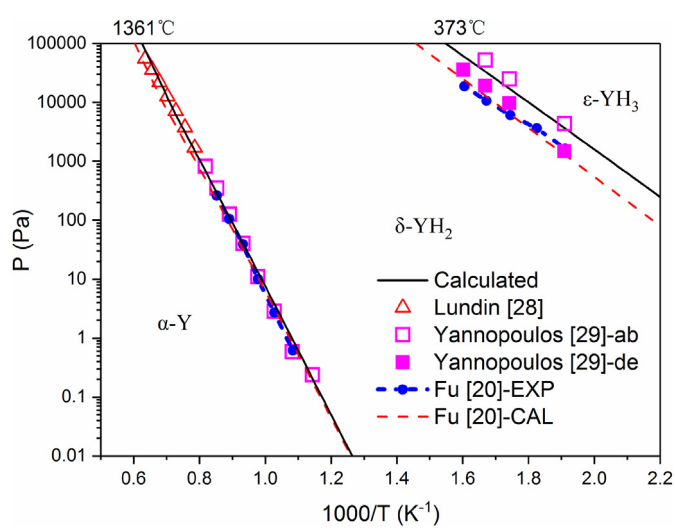

Fig. 4. The calculated van't Hoff plots of two-phase equilibria of α-Y+δ-YH_2 and δ-YH_2+ε-YH_3.

results obtained by Fu et al. [20] and the experimental results for the end-members of δ-YH_2 (fcc) and ε-YH_{3-x} (hcp) obtained by Akhachinskii [36].

5.2. Application in crack-free Zr–Y hydride fabrication

In our previous work, we demonstrated that yttrium is able to resist hydrogen loss from the zirconium hydride at the reactor working temperature by stabilizing the hydride matrix and the surface oxide layer at the same time [18]. Therefore the Zr–Y hydride can be a promising candidate for neutron moderator in compact nuclear reactors. However, during the hydrogenation of zirconium, the lattice distortion and volume expansion occur accompanied with phase transformation from metal to hydride, forming the brittle δ and ε zirconium hydride. Moreover, the equilibrium hydrogen pressures of zirconium and yttrium are significantly different. Upon hydrogenation, the volume expansion of zirconium hydride and yttrium hydride differs largely. Hence, it is a formidable challenge to fabricate the crack-free bulk Zr–Y hydride.

To fabricate the crack-free bulk Zr–Y hydride, three binary subsystems were combined and extrapolated to construct the Y–Zr–H ternary system. The Y–H system is assessed by this work. The Zr–H system was reported by Dupin et al. [52], Kőnigsberger et al. [53], Ukita et al. [54] and Yu and Macdonald [55]. This system consists of two allotropic solid solutions α-Zr and β-Zr, three non-stoichiometric hydrides γ-ZrH, δ-ZrH_2, and ε-ZrH_2, as well as the gas and liquid phase. The model assessed by Dupin et al. [52] is chosen for the extrapolation of ternary system because less optimized parameters was used. The γ-ZrH hydride is not included in our work since it is a metastable phase at low temperature. The Y–Zr system was optimized by Flandorfer et al. [56], He and Liu [57] and Bu et al. [58]. This system is composed of α-Zr, β-Zr, α-Y and β-Y, and fairly limited mutual solubility is found between yttrium and zirconium. The model parameters obtained by Bu et al. [58] is adopted in this work.

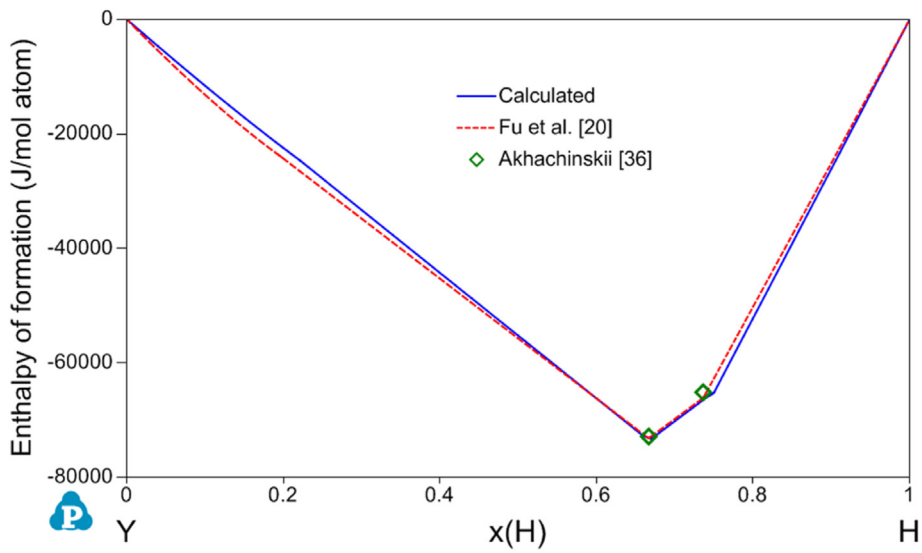

Fig. 5. The calculated enthalpies of formation at 25 °C as a function of hydrogen concentration.

For the Y–Zr–H ternary system, however, very few experimental data were reported. Isothermal curve was obtained for the Y–Zr–H system with a zirconium content of 44.4 at% at 700 °C [59]. No ternary compounds were found in the system. It is revealed that ZrH_x and YH_z show limited mutual solubility. This is consistent with our recent work, which demonstrated that little yttrium is dissolved as solid solution in the zirconium hydride. The PCI curve of $Zr_{0.444}Y_{0.556}$ at 700 °C was calculated using the Y–Zr–H system, as shown in Fig. 6. The different phase regions are separated by the vertical lines and denoted by the texts. The solid lines represent the calculated results, and the dashed lines represent the experimental results. The calculated initial and terminal points of the pressure plateaus deviate a little from the experimental results obtained by Fadeev et al. [59]. This difference may come from the measurement error and the lack of the experimental information of the ternary system for thermodynamic optimization.

With the aid of the assessed thermodynamic databases, the thermodynamic information, such as phase equilibrium relation and phase transition of the system, can be acquired and applied to establish the hydriding process of the Zr–Y alloys. Taking zirconium alloy with 20 at% Y as an example, according to the thermodynamic calculation results from the database, the δ-YH_x (x = 0.57) hydride phase starts to form under a hydrogen pressure of 423 Pa at 900 °C, but zirconium still exists in the form of β-Zr solid solution with a fairly low H/Zr ratio of 0.09, far lower than the solubility limits of hydrogen in β-Zr. The hydrogen content of β-Zr remains almost unchanged until all the α-Y is transformed to δ-YH_x (x = 1.14). The calculated PCI curve of $Zr_{0.8}Y_{0.2}$ at 900 °C is shown in Fig. 7. After the hydrogen pressure is increased to 1 bar, the H/Zr ratio of β-Zr gradually increases to 1.08, while the H/Y ratio reaches 1.94. If the temperature is then decreased, yttrium will be saturated with hydrogen and remain steady, and zirconium will undergo transformation from metal to hydride. Fig. 8 shows the calculated isothermal sections at 900 °C, 800 °C and 700 °C. When the temperature decreases under 1 bar of hydrogen pressure, the β-Zr metal gradually transforms to δ-ZrH_{2-x} and ε-ZrH_{2-x}. Very limited mutual solubilities between zirconium and yttrium, zirconium hydride and yttrium hydride were demonstrated. Because of the

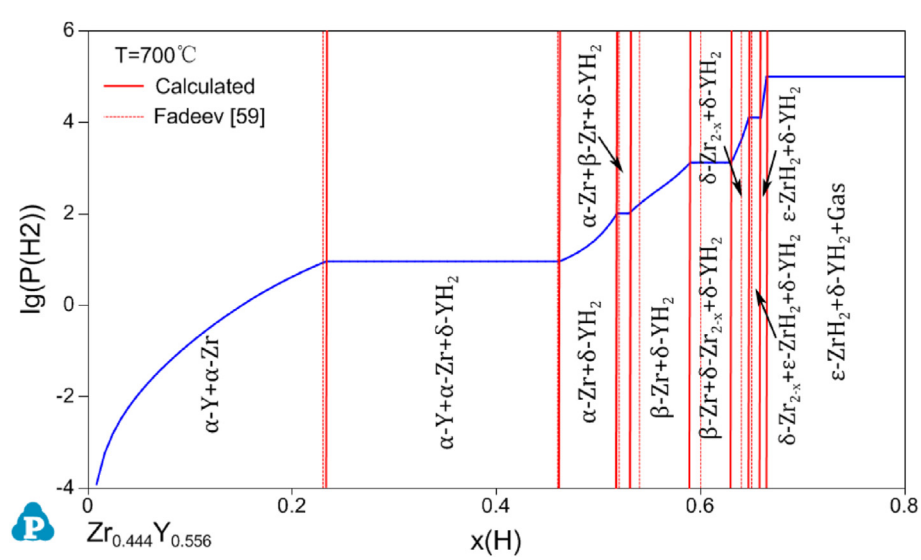

Fig. 6. The calculated PCI curve of $Zr_{0.444}Y_{0.556}$ at 700 °C.

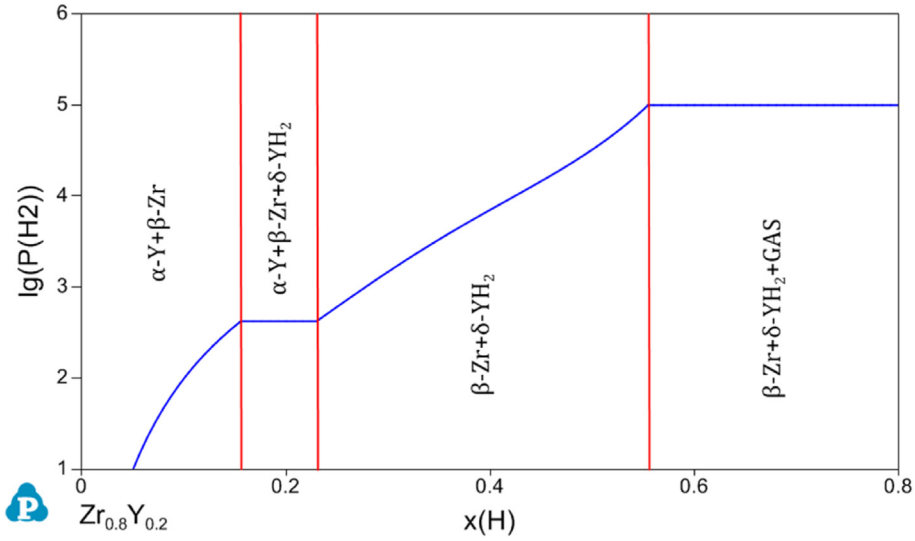

Fig. 7. The calculated PCI curve of $Zr_{0.8}Y_{0.2}$ at 900 °C.

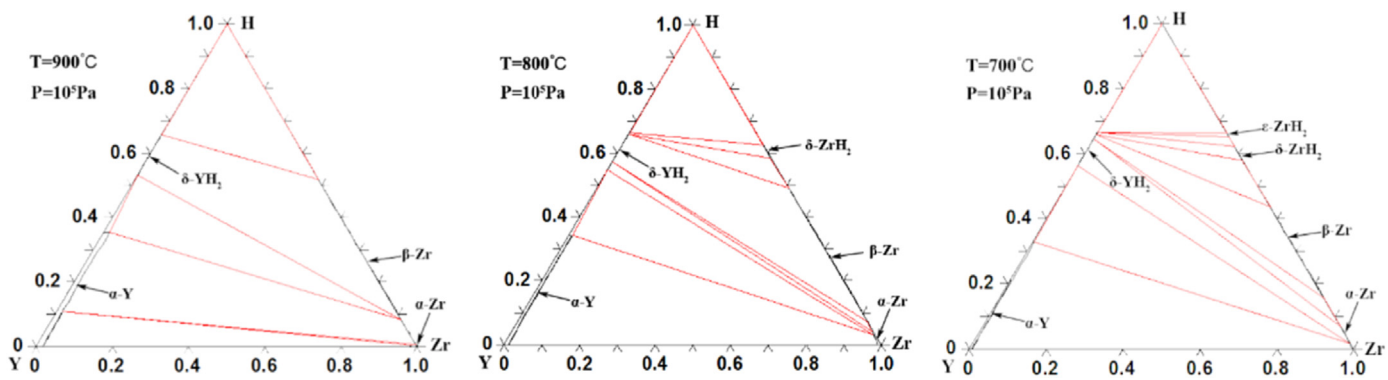

Fig. 8. The calculated isothermal sections at 900 °C, 800 °C and 700 °C.

limited mutual solubility between zirconium hydride and yttrium hydride, the thermodynamic calculation based on the separated Y–H and Zr–H binary systems is almost the same as the results of the Y–Zr–H ternary system. The Zr–Y alloy hydrides can be simply considered as the mixtures of zirconium hydride and yttrium hydride without any solid solution in them.

Based on the analysis above, the hydrogenation of Zr–Y alloys was carried out in three stages: isothermal hydriding, isobaric hydriding and terminal hydriding. The schematic diagrams of the hydriding process are presented in the calculated phase diagrams combined with isobars of Y–H and Zr–H system, as shown in the Figs. 3 and 9, respectively. Table 7 shows the detailed temperature and hydrogen pressure ranges, and the phase transformation occurring in the three hydriding stages. In the isothermal hydriding stage, the temperature was kept constant at 900 °C and the hydrogen pressure was increased from 0 to 1 bar. All the yttrium was transformed into δ-YH_{2-x}, and no δ zirconium hydride was formed in this stage. Subsequently, in the isobaric hydriding stage, the hydrogen pressure was kept at 1 bar constantly, and the furnace temperature was cooled down. The temperature cooling rate was controlled to be 1 °C/h, and the highest hydrogen flow rate was regulated to be 15 $mL \cdot min^{-1} \cdot mol^{-1}$. In this stage, zirconium in the alloys underwent the following transformation: β-Zr→δ-ZrH_{2-x}→ε-ZrH_{2-x}. When the temperature was decreasing, the H/Y and H/Zr ratio varied along the 1 bar isobar of Y–H and Zr–H system, respectively. The equilibrium temperature T* of the terminal hydriding stage was dependent on the final hydrogen content of the Zr–Y hydride and can be calculated according to the thermodynamic database. The hydrogen content of the Zr–Y hydride can thus be accurately controlled. With these three stages, the transformation from metal to hydride of yttrium and zirconium can be separated and carried out in different stages, so that the cracks formation during hydrogenation can be inhibited.

Zr–Y alloys with 1, 4, 10, 20 at% yttrium were arc-melted, which was described in detail elsewhere [60]. The as-received alloys were constituted by α-Zr and α-Y, and little yttrium was dissolved in zirconium after thermal treatment at 900 °C. The melted alloys were cut, mechanically polished, enveloped in molybdenum foil, and then hydrided in a self-assembled set-up, as shown in Fig. 10. The self-assembled hydriding setup mainly consists of electric tube furnace, vacuum system, pressure transmitter, mass flow controller and PID controller. The samples were first annealed at 900 °C in vacuum as low as 10^{-5} Pa for an hour in a stainless steel tube furnace. Then the high-purity hydrogen gas (5 N) was slowly admitted into the reaction chamber to make sure the hydrogen absorption of the alloys was conducted at a very low speed. The hydrogen flow rate was accurately controlled by a mass flow controller, and the hydrogen pressure was simultaneously monitored by a pressure transmitter. If the hydrogen pressure exceeded 1 bar, the hydrogen flow rate would be decreased to keep the

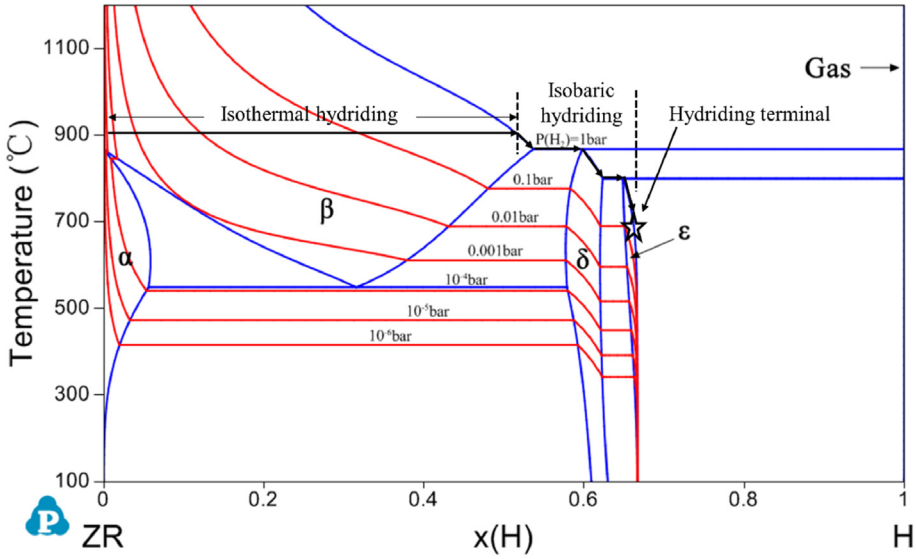

Fig. 9. The calculated phase diagram combined with isobars of the Zr–H system. The hydriding process is also denoted in the figure.

Table 7
The hydriding process of the Zr–Y hydride fabrication.

Hydriding stage	Temperature range/°C	Pressure range/bar	Phase transformation
Isothermal hydriding	900	0–1	α-Y → δ-YH_{2-x}
Isobaric hydriding	900-T*	1	β-Zr → δ-ZrH_{2-x} → ε-ZrH_{2-x}
Terminal hydriding	T*	1	–

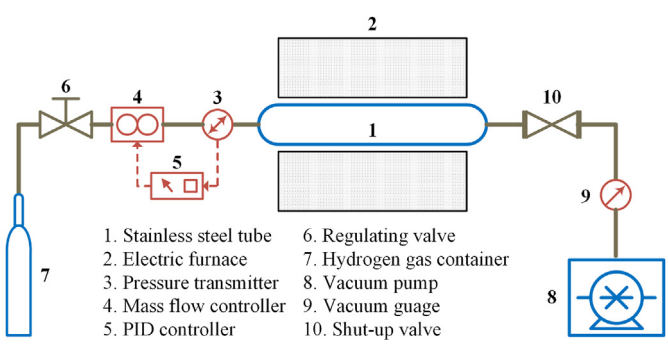

Fig. 10. The schematic diagram of the self-assembled hydriding set-up.

pressure constant using the PID controller. Upon hydriding, the hydrogen flow rate and the temperature cooling rate were kept at a considerably low level. By carefully controlling the furnace temperature and the hydrogen pressure, the hydriding process underwent isothermal hydriding, isobaric hydriding and terminal hydriding successively.

Using the designed hydrogenation process, a series of crack-free bulk Zr–Y hydrides containing 1, 4, 10, and 20 at% yttrium was successfully fabricated. The hydrogen content of the specimens was determined to be H/(Zr + Y) = 1.94 by measuring the weight before and after the hydrogenation. The as-prepared hydrides were mechanically polished and then etched using a mixed acid with volume ratio of HF:HNO_3:H_2O = 1:4:45. The metallographic microstructure of the samples was observed using optical microscope (Zeiss Axiovert 200 MAT).

Fig. 11 shows the metallographic microstructure of the Zr–Y hydrides with various Y contents. It can be found that a microcrack as long as 100 μm emerged in the sample of zirconium hydride with 1 at% Y. In this sample, the grain size was substantially large (approximately several hundreds of micrometers), and large numbers of parallel banded structures can be observed, which is the typical twin structure of ε zirconium hydride. This twin structure was formed by a martensitic transformation from the δ hydride phase, inducing large shear stress around ε hydride, which is blamed for the major reason for the cracks formation during hydrogenation [61,62]. It can be seen from the Fig. 11 that the grain size decreases sharply with the increase of the yttrium content. Yttrium can be considered as catalysts by providing "hydrogen-pump" effect, enabling a pinning effect to suppress the grain growth of zirconium hydride [63]. The strong grain refinement effect of yttrium on the zirconium hydride inhibits the formation of the banded twin structure, thus the stress induced by the martensitic transformation can be relaxed. Yttrium can also act as the nucleation sites, facilitating precipitation of high density uniformly distributed hydrides with small size, hence allowing a more uniform stress induced by the transformation from the metal to hydride [64,65]. Therefore, hydride embrittlement can be mitigated by yttrium, and no micro-cracks were observed in the other samples with larger Y content.

6. Conclusions

A self-consistent thermodynamic description of Y–H binary system is derived using the CALPHAD method with the aid of first-principle calculations. All the solid phases were described using only one three-sublattice model. The enthalpies of formation of the end-members calculated using the DFT method were adopted for the thermodynamic assessment. Only three model parameters were needed and optimized with the available experimental data from literature. The calculated phase equilibria and thermodynamic data agree well with experimental data.

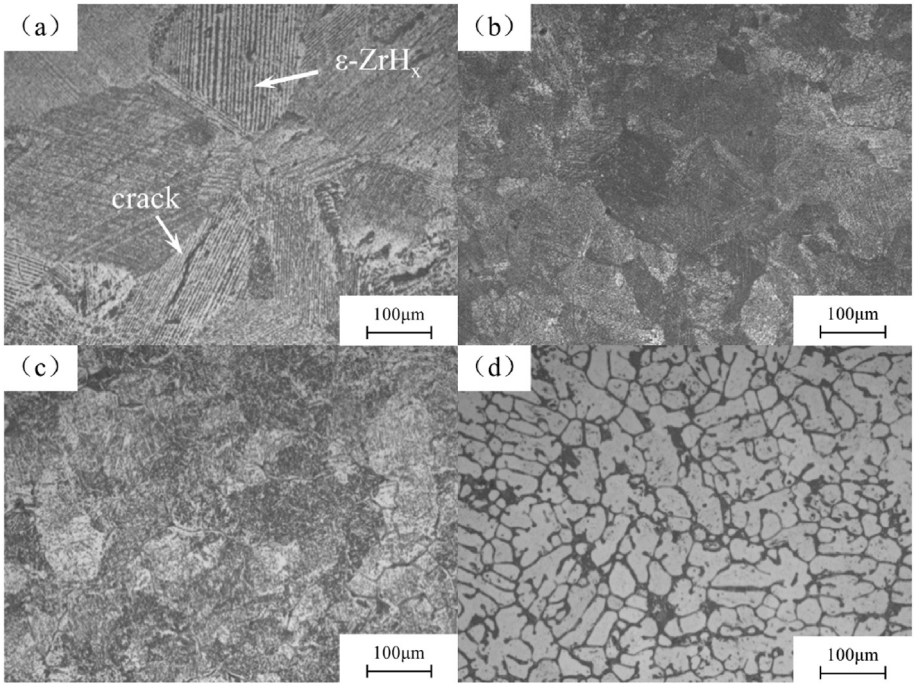

Fig. 11. The metallographic microstructure of the Zr–Y hydrides with various Y contents. (a) 1 at% Y; (b) 4 at% Y; (c) 10 at% Y; (d) 20 at% Y.

To fabricate the crack-free bulk Zr–Y hydride neutron moderator, the assessed Y–H binary system of this work, the Zr–H binary system adapted from literature and Y–Zr binary system were combined and extrapolated to construct the Y–Zr–H ternary system to design the hydriding process. The hydrogenation of Zr–Y alloys consists of three stages: isothermal hydriding, isobaric hydriding and terminal hydriding. By carefully controlling the temperature and hydrogen pressure, the transformation from metal to hydride of yttrium and zirconium can be separated and carried out in isothermal hydriding and isobaric hydriding, respectively, so that the cracks formation during hydrogenation can be inhibited. With the addition of yttrium, the banded twin structure of ε zirconium hydride formed by a martensitic transformation from the δ hydride phase can be retarded, and high density hydride with small size will be precipitated uniformly, thus allowing the stress induced by the phase transformation to be distributed uniformly and eventually relaxed. As a result, hydride embrittlement can be mitigated, and the bulk Zr–Y hydrides without micro-cracks can be fabricated.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time due to legal or ethical reasons.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiaqing Peng: Conceptualization, Methodology, Software, Writing - original draft. **Ming Wu:** Investigation, Validation, Formal analysis. **Fu Du:** Investigation, Validation, Formal analysis. **Fengli Yang:** Validation, Formal analysis. **Jianyun Shen:** Methodology, Writing - review & editing. **Lijun Wang:** Resources, Supervision, Funding acquisition. **Xinyu Ye:** Supervision, Project administration, Funding acquisition. **Guoqing Yan:** Conceptualization, Resources, Writing - original draft, Formal analysis.

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References

[1] J.M. Joubert, Thermodynamic modelling of metal–hydrogen systems using the Calphad method, J. Alloys Compd. 645 (2015) S379–S383.
[2] X.H. An, Q.F. Gu, J.Y. Zhang, S.L. Chen, X.B. Yu, Q. Li, Experimental investigation and thermodynamic reassessment of La–Ni and $LaNi_5$–H systems, Calphad 40 (2013) 48–55.
[3] Q. Luo, S.L. Chen, J.Y. Zhang, L. Li, K.C. Chou, Q. Li, Experimental investigation and thermodynamic assessment of Nd–H and Nd–Ni–H systems, Calphad 51 (2015) 282–291.
[4] Q. Li, Q. Luo, Q.F. Gu, Insights into the composition exploration of novel hydrogen storage alloys: evaluation of the Mg–Ni–Nd–H phase diagram, J. Mater. Chem. 5 (2017) 3848–3864.
[5] J.N. Huiberts, R. Griessen, J.H. Rector, R.J. Wijngaarden, J.P. Dekker, D.G. Groot, N.J. Koeman, Yttrium and lanthanum hydride films with switchable optical properties, Nature 380 (1996) 231.
[6] K. Fu, G. Li, J. Li, Y. Liu, W. Tian, J. Zheng, X. Li, Study on the thermodynamics of the gadolinium-hydrogen binary system (H/Gd=0.0–2.0) and implications to metallic gadolinium purification, J. Alloys Compd. 673 (2016) 131–137.
[7] K. Fu, G. Li, J. Li, Y. Liu, W. Tian, X. Li, Experimental study and thermodynamic assessment of the dysprosium-hydrogen binary system, J. Alloys Compd. 696 (2017) 60–66.
[8] A. Mascaro, C. Toffolon-Masclet, C. Raepsaet, J.C. Crivello, J.M. Joubert, Experimental study and thermodynamic description of the erbium–hydrogen–zirconium ternary system, J. Nucl. Mater. 456 (2015) 7–16.
[9] T. Yang, Q. Li, N. Liu, C. Liang, F. Yin, Y. Zhang, Improved hydrogen absorption and desorption kinetics of magnesium-based alloy via addition of yttrium,

J. Power Sources 378 (2018) 636–645.
[10] Y. Guo, Y. Li, B. Liu, W. Liu, X. Liang, Q. Gu, Q. Li, Precipitation mechanism of Mg_2Ni in Mg-Ni-Y studied by STEM, 3DAP and first-principles calculations, J. Alloys Compd. 750 (2018) 117–123.
[11] X. Yang, S. Wu, S. Lü, L. Hao, X. Fang, Effects of Ni levels on microstructure and mechanical properties of Mg-Ni-Y alloy reinforced with LPSO structure, J. Alloys Compd. 726 (2017) 276–283.
[12] P. Ngene, A. Longo, L. Mooij, W. Bras, B. Dam, Metal-hydrogen systems with an exceptionally large and tunable thermodynamic destabilization, Nat. Commun. 8 (2017) 1846.
[13] P. Ngene, T. Radeva, M. Slaman, R.J. Westerwaal, H. Schreuders, B. Dam, Seeing Hydrogen in Colors: low-cost and highly sensitive eye readable hydrogen detectors, Adv. Funct. Mater. 24 (2014) 2374–2382.
[14] M. La, N. Li, R. Sha, S. Bao, P. Jin, Excellent photochromic properties of an oxygen-containing yttrium hydride coated with tungsten oxide (YH_x:O/WO_3), Scripta Mater. 142 (2018) 36–40.
[15] T. Mongstad, C. Platzer-Björkman, J.P. Maehlen, L.P.A. Mooij, Y. Pivak, B. Dam, E.S. Marstein, B.C. Hauback, S.Z. Karazhanov, A new thin film photochromic material: oxygen-containing yttrium hydride, Sol. Energy Mater. Sol. Cell. 95 (2011) 3596–3599.
[16] M. Ito, J. Matsunaga, D. Setoyama, H. Muta, K. Kurosaki, M. Uno, S. Yamanaka, Thermal properties of yttrium hydride, J. Nucl. Mater. 344 (2005) 295–297.
[17] R. Van Houten, Selected engineering and fabrication aspects of nuclear metal hydrides (Li, Ti, Zr, and Y), Nucl. Eng. Des. 31 (1974) 434–448.
[18] J.Q. Peng, Y. Chen, M. Wu, G.Q. Yan, L.J. Wang, J.S. Li, Z.C. Guo, Hydrogen desorption behavior of the hydrides of Zr-Y alloys under Ar and CO_2 atmosphere, J. Alloys Compd. 693 (2017) 103–109.
[19] M. Wu, J.Q. Peng, G.Q. Yan, Y. Chen, Z. He, Z.G. Xu, L.J. Wang, Preparation and properties of composite hydrogen permeation barrier on $ZrH_{1.8}$ by sol–gel technique, Surf. Coating. Technol. 352 (2018) 159–165.
[20] K. Fu, X. Jiang, Y. Guo, S. Li, J. Zheng, W. Tian, X. Li, Experimental investigation and thermodynamic assessment of the yttrium-hydrogen binary system, Prog. Nat. Sci.: Mater. Int. 28 (2018) 332–336.
[21] D. Khatamian, F.D. Manchester, The H–Y (Hydrogen-Yttrium) system, J. Phase Equil. 9 (1988) 252–260.
[22] D. Khatamian, C. Stassis, B.J. Beaudry, Location of deuterium in α-yttrium, Phys. Rev. B 23 (1981) 624–627.
[23] B.J. Beaudry, F.H. Spedding, The solubility of RH_{2-x} in Gd, Er, Tm, Lu and Y from ambient to 850°C, Metall. Mater. Trans. B 6 (1975) 419–427.
[24] J.H. Weaver, R. Rosei, D.T. Peterson, Crystal structure of $YD_{1.96}$ and $YH_{1.98}$ by neutron diffraction, J. Less Common Met. 21 (1980) 2622–2624.
[25] M. Mansmann, W.E. Wallace, The structure of HoD_3, J. Phys. 25 (1964) 454–459.
[26] L.C. Beavis, Characteristics of some binary transition metal hydrides, J. Less Common Met. 19 (1969) 315–328.
[27] A. Pebler, W.E. Wallace, Crystal structures of some lanthanide hydrides, J. Phys. Chem. 66 (1962) 148–151.
[28] C.E. Lundin, J.P. Blackledge, Pressure-temperature-composition relationships of the yttrium-hydrogen system, J. Electrochem. Soc. 109 (1962) 838–842.
[29] L.N. Yannopoulos, R.K. Edwards, P.G. Wahlbeck, The thermodynamics of the yttrium-hydrogen system, J. Phys. Chem. 69 (1965) 2510–2515.
[30] G.M. Begun, J.F. Land, J.T. Bell, High temperature equilibrium measurements of the yttrium–hydrogen isotope (H_2, D_2, T_2) systems, J. Chem. Phys. 72 (1980) 2959–2966.
[31] M. Tanase, P.W. Fisher, J.S. Watson, Isothermal equilibrium pressures of a Y-Nb alloy-H_2 system, J. Less Common Met. 99 (1984) 23–31.
[32] A.S. Chernikov, V.I. Savin, V.N. Fadeev, N.A. Landin, L.A. Izhvanov, Thermodynamic and physical properties of yttrium and some rare earth hydrides, J. Less Common Met. 130 (1987) 441–452.
[33] J.E. Bonnet, C. Juckum, A. Lucasson, Solid solutions of H and D in yttrium metal, J. Phys. F Met. Phys. 12 (1982) 699–711.
[34] D.L. Anderson, R.G. Barnes, S.O. Nelson, D.R. Torgeson, Wide-line proton magnetic resonance in the solid solution phase of the yttrium-hydrogen system, Phys. Lett. 74 (1979) 427–431.
[35] P. Dantzer, O.J. Kleppa, High temperature thermodynamics of the yttrium–hydrogen systems, J. Chem. Phys. 73 (1980) 5259–5263.
[36] V.V. Akhachinskii, Enthalpy of formation of YH_2 and $YH_{2.79}$, Proc. Int. Symp. Thermodyn. Nucl. Mater. (1980) 155–160.
[37] B. Sundman, J. Ågren, A regular solution model for phases with several components and sublattices, suitable for computer applications, J. Phys. Chem. Solid. 42 (1981) 297–301.
[38] A.T. Dinsdale, SGTE data for pure elements, Calphad 15 (1991) 317–425.
[39] J.T. Udovic, Q. Huang, A. Santoro, J.J. Rush, The nature of deuterium arrangements in YD_3 and other rare-earth trideuterides, Z. Kristallogr. (2008) 697.
[40] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys. Rev. B 54 (1996) 11169–11186.
[41] G. Kresse, From ultrasoft pseudopotentials to the projector augmented-wave method, Phys. Rev. B 59 (1999) 1758–1775.
[42] J.P. Perdew, M. Ernzerhof, K. Burke, Rationale for mixing exact exchange with density functional approximations, J. Chem. Phys. 105 (1996) 9982–9985.
[43] P.E. Blöchl, Projector augmented-wave method, Phys. Rev. B 50 (1994) 17953–17979.
[44] S.L. Shang, Y. Wang, D. Kim, Z.K. Liu, First-principles thermodynamics from phonon and Debye model: application to Ni and Ni_3Al, Comput. Mater. Sci. 47 (2010) 1040–1048.
[45] T. Chihi, M. Fatmi, A. Bouhemadou, Structural, mechanical and electronic properties of transition metal hydrides MH_2 (M = Ti, Zr, Hf, Sc, Y, La, V and Cr), Solid State Sci. 14 (2012) 583–586.
[46] S. Kanagaprabha, A.T. Asvinimeenaatci, G. Sudhapriyanga, A. Jemmycinthia, R. Rajeswarapalanichamy, K. Iyakutti, First principles study of stability and electronic structure of TMH and TMH_2 (TM = Y, Zr, Nb), Acta Phys. Pol., A 123 (2013) 126–131.
[47] L. Zhenli, C. Xinlu, Mechanical and thermodynamic properties of cubic YH_2 under high pressure: prediction from first-principles study, Chin. Phys. B 23 (2014) 46201.
[48] G. Kresse, Dissociation and sticking of H_2 on the Ni(111), (100), and (110) substrate, Phys. Rev. B 62 (2000) 8295–8305.
[49] J. Andersson, T. Helander, L. Hoglund, P. Shi, B. Sundman, THERMO-CALC & DICTRA, computational tools for materials science, Calphad 26 (2002) 273–312.
[50] J.M. Joubert, CALPHAD modeling of metal–hydrogen systems: a review, J. Occup. Med. 64 (2012) 1438–1447.
[51] S. Chen, W. Cao, C. Zhang, J. Zhu, F. Zhang, Q. Li, J. Zhang, Calculation of property contour diagrams, Calphad 55 (2016) 63–68.
[52] N. Dupin, I. Ansara, C. Servant, C. Toffolon, C. Lemaignan, J. Brachet, A thermodynamic database for zirconium alloys, J. Nucl. Mater. 275 (1999) 287–295.
[53] E. Konigsberger, G. Eriksson, W.A. Oates, Optimisation of the thermodynamic properties of the Ti–H and Zr–H systems, J. Alloys Compd. 299 (2000) 148–152.
[54] S. Ukita, H. Ohtani, M. Hasebe, Thermodynamic analysis of the Ti-H and Zr-H binary phase diagrams, J. Jpn. Inst. Metals 71 (2007) 721–729.
[55] Z. Yu, D.D. Macdonald, Thermodynamics of the Zr-H binary system related to nuclear fuel sheathing and pressure tube hydriding, J. Nucl. Mater. 423 (2012) 87–92.
[56] H. Flandorfer, J. Grobner, A. Stamou, N.D. Hassiotis, A. Saccone, P. Rogl, R. Wouters, H. Seifert, D. Maccio, R. Ferro, Experimental investigation and thermodynamic calculation of the ternary system Mn-Y-Zr, Zeitschrift Fur Metallkunde 88 (1997) 529–538.
[57] X. He, H.S. Liu, Determination of the isothermal section of the Cu–Zr–Y ternary system at 978K, J. Alloys Compd. 475 (2009) 245–251.
[58] M. Bu, P. Wang, H. Xu, S. Liu, C. Sha, Y. Du, F. Pan, A. Tang, Experimental investigation and thermodynamic modeling of the Zr-Y system, J. Min. Metall. B Metall. 46 (2010) 181–192.
[59] V. Fadeev, L. Eremina, L. Izhvanov, Phase Equilibria in the Y-Zr-H System, Izvestiya Akademii Nauk SSSR, Metally, 1977, pp. 234–238.
[60] J.Q. Peng, Y. Chen, G.Q. Yan, M. Wu, L.J. Wang, J.S. Li, Solid solubility extension and microstructure evolution of cast zirconium yttrium alloy, Rare Met. 35 (2016) 325–330.
[61] C. Zhao, X. Song, Y. Yang, B. Zhang, Hydrogen absorption cracking of zirconium alloy in the application of nuclear industry, Int. J. Hydrogen Energy 38 (2013) 10903–10911.
[62] A. Chernikov, V. Syasin, V. Kostin, E. Boiko, Influence of hydrogen content on the strength and the presence of defects in ε-zirconium hydride, J. Alloys Compd. 330 (2002) 393–395.
[63] Q. Li, Y. Li, B. Liu, X. Lu, T. Zhang, Q. Gu, The cycling stability of the in situ formed Mg-based nanocomposite catalyzed by YH_2, J. Mater. Chem. 5 (2017) 17532–17543.
[64] L. Chen, X. Wang, W. Gong, H. Zhang, Effect of yttrium addition on microstructure and orientation of hydride precipitation in Zr-1Nb alloy, Int. J. Hydrogen Energy 39 (2014) 21116–21126.
[65] I. Batra, R. Singh, P. Sengupta, B. Maji, K. Madangopal, K. Manikrishna, R. Tewari, G. Dey, Mitigation of hydride embrittlement of zirconium by yttrium, J. Nucl. Mater. 389 (2009) 500–503.